

FOAMI: Enhancing ICS Threat Detection via Feature Optimization, Realistic Augmentation, and Mutual Inference

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Abstract. Industrial Control Systems (ICS) are now encountering a multitude of advanced cyber dangers, particularly those associated with the Internet. However, the present study continues to delineate several research gaps, including the restricted sample size, the subpar quality of the training data, and the inability to capitalize on the advantages of various Artificial Intelligence (AI) models. This study introduces a novel ICS threat detection system, named FOAMI (Feature Optimization, Realistic Data Augmentation, and Mutual Inference), which incorporates feature optimization, realistic data augmentation techniques, and mutual inference. Our FOAMI facilitates superior feature performance, increased layer separation by targeted aggregation, and overall learning enhancement via mutual inference. Extensive experimental findings on the standard industrial dataset IEC 60870-5-104 indicate that our approach significantly enhances detection accuracy, attaining a detection rate of 89.00% and decreasing the false alarm rate to only 0.99%, outperforming state-of-the-art sophisticated approaches.

Keywords: ICS Cybersecurity, Threat Detection, Covariate Shift, Feature Selection, Data Augmentation, Mutual Inference

1 Introduction

Information systems that support critical domains such as energy, healthcare, manufacturing, and transportation are fundamental to modern industry. By bridging physical and digital operations, they enable automation, real-time optimization, continuous monitoring, and enforcement of security policies. Nonetheless, the growing adoption of internet-connected ICS, particularly Supervisory Control and Data Acquisition (SCADA)-based systems, has simultaneously improved efficiency and expanded the potential attack surface [17]. Widely used protocols like IEC 60870-5-104 and DNP3, essential for smart grids and industrial control, facilitate reliable data exchange but remain susceptible to Man-in-the-Middle (MITM) and Distributed Denial of Service (DDoS) attacks because of limited

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access control and weak authentication mechanisms [13]. This vulnerability can lead to severe operational disruptions, economic losses, and even safety risks to human life. On the one hand, traditional cybersecurity methods struggle to protect ICS environments due to the demand for real-time response as well as the unique features of industrial protocols. On the other hand, machine learning (ML) and deep learning (DL) models can learn from normal network operational data and consequently detect anomalies and malicious activities without relying on predefined rules. Therefore, developing an AI-based Intrusion Detection System (IDS) has become a top priority to effectively protect critical infrastructure.

Despite their strong potential, ML and DL models still face numerous challenges in the ICS environment. Many feature engineering methods rely on specific protocols (e.g., Modbus, DNP3), which limits scalability and shows the need for protocol-independent features. Existing datasets [17] have quality problems and do not fully reflect real-world cases, while ICS traffic is highly imbalanced, making rare attacks hard to detect. In addition, advanced data augmentation methods such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), although effective for general IDS datasets (e.g., NSL-KDD, CIC-IDS 2017) [9], are not well explored in ICS. Finally, hybrid models that combine Gradient Boosted Trees (GBT) with DL are still limited, which reduces the robustness of detection systems. To overcome the challenges mentioned above, this research aims to strengthen both the effectiveness and robustness of threat detection mechanisms in ICS. The core research questions are:

- **RQ1:** How can the quality and discriminability of training data be improved to enhance the recognition of diverse cyberattacks and mitigate covariate shift in ICS environments?
- **RQ2:** Does generating additional synthetic data for attack types that are frequently confused improve the recognition performance for those attacks?
- **RQ3:** What advanced ensemble or machine learning strategies can be developed to improve intrusion detection accuracy while ensuring long-term adaptability against evolving ICS threats?

To answer these questions, we propose a novel intrusion detection framework with three main contributions:

- We present the first systematic evaluation of advanced data augmentation techniques designed for ICS threat detection.
- We introduce a hybrid ensemble engine that combines GBT and DL models through Mutual Inference (MI).
- We provide comprehensive validation on IEC-104 protocol, showing improved robustness and higher detection accuracy.

2 Related Work

2.1 AI-powered Threat Detection in ICS

Most existing studies mainly use general Internet of Things (IoT) or traditional network datasets, such as NSL-KDD, CIC-IDS 2018, and UNSW-NB-15

[25,24,11,12,7], which differ in characteristics and attack scenarios from ICS-specific datasets like IEC 104. The unique properties of ICS protocols, including specialized packet structures and strict operational contexts, require customized detection methods. Therefore, this work focuses on using ICS datasets and proposes feature selection and data augmentation methods designed for ICS network traffic. The goal is to address current gaps and provide new contributions to the field.

Radoglou-Grammatikis et al. [19] studied threats to IEC 104, including unauthorized access, man-in-the-middle attacks, and DDoS, highlighting weaknesses caused by poor authentication in ICS/SCADA systems. Study [1] proposed an intrusion detection system that combines traffic monitoring with anomaly detection, while [17] used decision trees and SDN to build the IEC-104 Intrusion Detection Dataset (IEC-104-IDS). Asimopoulos et al. [3] tested classifiers such as XGBoost, Random Forest, and MLP, reporting 82.44% accuracy and an F1-score of 80.98%.

2.2 Feature Optimization

Feature importance analysis is essential for interpreting machine learning models because it highlights the features that have the greatest impact on predictions. A common method used for this task is SHapley Additive exPlanations (SHAP), which has been applied in many studies to gain a deeper understanding of AI models [21] and to improve system performance by reducing feature dimensionality [4]. Based on Shapley values from game theory, SHAP provides a reliable way to effectively measure how much each feature contributes to both individual predictions as well as the overall model output. In this research, we applied a thresholding value of 0.0005 on the ExtraTrees model (the best-performing baseline) to eliminate features with low contribution. This threshold was carefully selected to ensure that, after retraining, all performance metrics (Accuracy, Precision, Recall, and F1-score) remained equal to or higher than their original values, while the False Alarm Rate (FAR) specifically decreased. The final dataset is then reconstructed using these consistently selected significant features, which subsequently facilitates more efficient and effective training and inference in ML/DL models.

2.3 Data Augmentation

Data augmentation is widely used in ICS intrusion detection to expand training data and improve robustness under covariate shift. Traditional methods such as Synthetic Minority Over-sampling Technique (SMOTE) and Adaptive Synthetic Sampling (ADASYN) generate new samples by interpolation [5], but they often fail to capture non-linear dependencies and may create unrealistic protocol records. Deep generative models, including GAN-based and VAE-based approaches, can learn the joint feature distribution, with Conditional Tabular GAN (CTGAN) and Tabular Variational Autoencoder (TVAE) [26] improving performance on ICS datasets [2,14], and GAN-based synthesis enhancing anomaly detection on process data such as SWaT and HAI [6]. More recently, diffusion models like

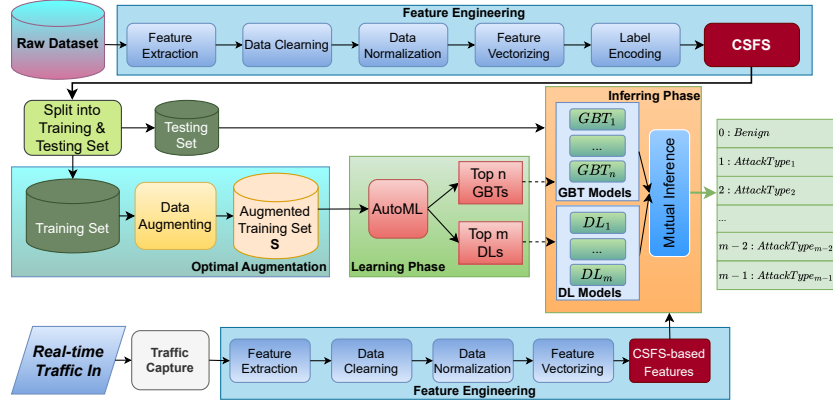


Fig. 1. Architecture of FOAMI-powered Threat Detection.

TabDDPM [10] and TabDiff [8] have produced higher-quality and more stable samples than GANs, which is useful under shifting ICS traffic. In parallel, Large Language Model (LLM)-based methods such as GReaT [23] and CLLM [15] have been applied to tabular synthesis, showing potential for modeling protocol semantics and rare attack patterns, though their use in ICS is still at an early stage.

3 Proposed Method

3.1 Approach Direction

Intrusion detection in ICS faces two major obstacles. On one hand, covariate shift alters the distribution of features between training and deployment, leading to a decline in predictive reliability. On the other, balanced datasets still contain classes that are difficult to distinguish because their feature patterns overlap extensively. These issues require a framework that is both resilient to distributional drift and capable of enlarging subtle distinctions between attacks.

To overcome these challenges, we follow three design principles. First, shift-aware optimization is applied to keep only the features that remain stable during the change in data distributions. Second, targeted data augmentation is utilized to create synthetic samples, which improves the separation between attack classes that are often confused. Finally, we design a hybrid ensemble with mutual inference, where GBT and DL models are fused in an adaptive manner, allowing them to reinforce rather than merely average each other's predictions. Together, these principles define our framework, namely **FOAMI**, short for the ICS threat detection by employing **F**eature **O**ptimization, **R**ealistic Data **A**ugmentation and **M**utual **I**nference.

Figure 1 illustrates the FOAMI pipeline, which is designed with four tightly connected phases to ensure adaptability, scalability, and accurate inference in ICS environments: (i) **Feature Engineering**: The input, either a raw dataset or real-time traffic, undergoes a comprehensive preprocessing workflow before

model training. The CSFS module serves as the core of this phase, responsible for filtering and retaining only the most stable features under distributional shift. **(ii) Optimal Augmentation Phase:** After identifying rare or easily confused attack classes, selective synthetic data generation is applied. The results of different generation models are compared to determine the most effective one. This process increases both the density and diversity of minority-class samples while expanding the decision boundaries between attack categories. **(iii) Learning Phase:** An AutoML procedure is employed to automatically select and train the most suitable model architectures for the given dataset. **(iv) Inferring Phase:** Two model families, GBT and DL, are trained in parallel to establish the foundation for model fusion. The *MI* module adaptively integrates predictions from both branches, enabling dynamic weighting between models and thus improving bias reduction and stability in intrusion detection outcomes.

3.2 CSFS-powered Feature Optimization

The foundation of our approach is the Covariate Shift Feature Selection (CSFS) framework [16], which addresses covariate shift by filtering unstable features and selecting the most informative ones through SHAP analysis. This ensures that the learning process is not undermined by distributional drift. The CSFS framework improves dataset robustness through two main stages: covariate shift adaptation and SHAP-based selection. First, covariate shift adaptation is performed: histogram overlap and Kolmogorov–Smirnov (KS) tests highlight unstable features, discriminative distance quantifies how easily training and testing data can be separated, PCA visualization provides further validation, and unstable features identified are then removed from the dataset. Second, the remaining features undergo SHAP-based selection, where multiple models evaluate their contributions. Only those with consistently high SHAP values are retained, ensuring the final feature set is both reliable and interpretable.

Building on this base, we develop the FOAMI framework, which improves performance in two critical ways. First, we add data augmentation using several SoTA models such as TVAE, GPT2, etc. Samples are selectively generated for attack classes that are most difficult to distinguish. This step reduces class confusion that feature selection alone cannot resolve. Second, we design a hybrid ensemble with Mutual Inference, where GBT and DL models exchange complementary signals, allowing reliable predictions to be amplified and uncertain ones suppressed. With these enhancements, FOAMI demonstrates a clear improvement over CSFS. While CSFS already raises F1 scores to around 85%, the augmented and ensemble-based design of FOAMI pushes performance close to 89%, accompanied by a lower false alarm rate.

3.3 Realistic Data Augmentation Strategy

To improve the quality of the training dataset, especially when the number of samples in each class is small, such as in the IEC-104 dataset, a strategy for generating additional samples is needed. Our approach to this challenge is based on evaluating SoTA data synthesis models, including (i) feature space-based and local interpolation (such as SMOTE), multivariate normal distribution

(e.g., GaussianCopula), conditional distributions and data generation (CTGAN, CopulaGAN), diffusion for tabular data (TabDiff), and LLM-based augmentation. Each method will be trained on the original dataset and generate new data, which will then be evaluated based on quantitative criteria such as distribution similarity (Wasserstein distance, KS-test), downstream model performance (F1-score, AUC), and semantic validity according to the operational characteristics of the network protocols used in ICS, such as IEC-104. The augmentation techniques employed in our study include classical oversampling, deep generative, diffusion-based, and LLM-based methods, all applied to the ICS dataset for synthetic data generation.

Classical Methods We used SMOTE [5], which creates new samples by interpolating between neighboring feature vectors. It was implemented via the `imbalanced-learn`¹ library to expand the training data.

Deep Generative Models Using the `SDV`² library, we trained CTGAN, CopulaGAN, GaussianCopula, and TVAE [26], which generate rows by modeling the joint distribution of tabular data. CTGAN relies on adversarial training, TVAE on variational autoencoding, while Copula-based models combine marginal distribution fitting with generative sampling.

Diffusion-based Models We adopted TabDiff [8], which applies iterative denoising with feature-wise noise schedules and a transformer-based denoiser to generate high-fidelity samples. The official implementation was used for training and synthesis on our dataset.

LLM-based Method Generating realistic tabular data is challenging in specialized domains, as it often requires domain knowledge beyond raw statistics. To address this, we use the GReaT framework [23] with GPT-2 [18], which encodes table rows as text, fine-tunes GPT-2 on these sequences, and decodes the output back into tabular form. This process captures feature dependencies and mimics how ICS experts describe anomalies, yielding synthetic data that is both valid and useful for model training.

3.4 GBT+DL Learning and Mutual Inference

To improve the performance of detecting ICS intrusion attack risks and address RQ3, we propose leveraging multiple ML/DL models and combining them. Based on actual experimental results, we introduce a mutual inference mechanism between two groups of ML models, GBT and DL, in order to improve accuracy in multi-class classification. Instead of simply combining the outputs of the models, this mechanism uses internal consensus among similar models to adjust the inference weights, thereby generating a final prediction with higher confidence. Essentially, mutual inference is based on the following key ideas:

1. Average Prediction: For each sample x_i , calculate the average output of the GBT and DL models, respectively.
2. Measure internal consensus: Calculate the cosine similarity between each model and the group average to measure the level of consensus: $A_{GBT}(i), A_{DL}(i)$.

¹ <https://imbalanced-learn.org>

² <https://sdv.dev>

3. Standardize agreement: Bring it to the same scale by dividing by the largest value in the set.
4. Calculate the final inference weight: The new weights are adjusted based on the average consensus and the coefficients α and β , combined with the initial weights.
5. Final prediction: Calculate the weighted sum of the average outputs of GBT and DL, then select the class with the highest probability as the predicted label $\hat{y}_i$ and record the confidence c_i as the highest probability.

Algorithm 1: Mutual Inference Ensemble Prediction

Input: Input samples $X = \{x_1, x_2, \dots, x_N\}$; List of n GBT models $\mathcal{M}_{GBT} = \{m_1, \dots, m_n\}$; List of m DL models $\mathcal{M}_{DL} = \{d_1, \dots, d_m\}$; Number of classes n_{class} ; Baseline weights $w_{GBT}^{base}, w_{DL}^{base}$; Confidence scaling factors α, β

1. **Output:** Predicted labels $\hat{Y} = \{\hat{y}_1, \dots, \hat{y}_N\}$; Confidence scores $C = \{c_1, \dots, c_N\}$

2 Initialize predictions list: $\hat{Y} \leftarrow []$, confidences list: $C \leftarrow []$;

3 **for** each sample index $i \in \{1, \dots, N\}$ **do**

// Extract predictions for sample i :

4 $p_{GBT}^i = [p_{m_1}(x_i), \dots, p_{m_n}(x_i)]$

5 $p_{DL}^i = [p_{d_1}(x_i), \dots, p_{d_m}(x_i)]$

// Compute confidence scores:

6 $c_{GBT}^i(k) = \max(p_{GBT}^i(k)) - \text{second_max}(p_{GBT}^i(k))$, for $k = 1, \dots, n$

7 $c_{DL}^i(l) = 1 - \frac{-\sum p_{DL}^i(l) \log p_{DL}^i(l)}{\log(n_{class})}$, for $l = 1, \dots, m$

8 Initialize agreement matrix $A \in \mathbb{R}^{n \times m}$;

9 **for** $k = 1$ **to** n **do**

10 **for** $l = 1$ **to** m **do**

11 $A_{k,l} = \text{cosine_similarity}(p_{GBT}^i(k), p_{DL}^i(l))$;

12 **end**

13 **end**

// Compute raw weights:

14 $w_{GBT}^{raw} = \alpha \times c_{GBT}^i \times \text{mean over row}(A) + (1 - \alpha)w_{GBT}^{base}$

15 $w_{DL}^{raw} = \beta \times c_{DL}^i \times \text{mean over column}(A) + (1 - \beta)w_{DL}^{base}$

// Normalize weights:

16 $w_{GBT} = \frac{w_{GBT}^{raw}}{\sum w_{GBT}^{raw}}, \quad w_{DL} = \frac{w_{DL}^{raw}}{\sum w_{DL}^{raw}};$

// Aggregate ensemble prediction:

17 $p_{ensemble}^i = \sum_{k=1}^n w_{GBT}(k) \times p_{GBT}^i(k) + \sum_{l=1}^m w_{DL}(l) \times p_{DL}^i(l)$;

// Predict label and confidence:

18 $\hat{y}^i = \arg \max p_{ensemble}^i, \quad c^i = \max p_{ensemble}^i$

// Append predictions:

19 $\hat{Y}.append(\hat{y}^i), \quad C.append(c^i)$

20 **end**

21 **return** $\hat{Y}, C$

In the MI mechanism, the two hyperparameters α and β act as dynamic weight regulators, which help determine the level of contribution between the two groups of GBT and DL models during the ensemble inference process. The introduction

of these two parameters stems from the differences between the two model groups: GBT has advantages in stability, explainability, and reliability on linear or well-engineered features; while DL excels in deep nonlinear representation, learning complex relationships within data, but is more susceptible to noise and overfitting. Therefore, α and β are designed to balance these two characteristics to create a more flexible fusion mechanism.

At initialization, the base weights of the two model groups are set to a ratio of 60:40 (GBT:DL), reflecting the higher reliability and interpretability of tree-based learners in industrial network traffic. These initial weights are then dynamically updated according to each group’s internal consensus level, ensuring that the more stable and coherent group gains greater influence in each specific inference. This adaptive adjustment allows the ensemble to emphasize robustness when models agree strongly and flexibility when deep models capture new attack patterns.

During the inference process, α adjusts the influence weight of the GBT group, while β controls the contribution of the DL group. This mechanism is architecture-agnostic and can be applied to any set of models. The overall process is abstracted in Algorithm 1.

The choice of cosine similarity in the weighting mechanism is motivated by its ability to measure directional agreement between model outputs rather than raw magnitude. In multi-class intrusion detection, two models can produce probabilities of different scales yet still rank classes in a similar order. Cosine similarity effectively captures this shared orientation, making it a robust indicator of inter-model consensus. When used for weighting, it strengthens agreement-driven decisions while preventing overreliance on poorly calibrated models.

From a statistical viewpoint, cosine similarity behaves like a normalized form of correlation, bounded between -1 and 1 , allowing stable estimation of coherence among diverse learners. Integrating this metric into the mutual inference process helps reduce ensemble variance while still preserving the diversity that ensemble learning relies on to generalize effectively. Consequently, cosine-based weighting enables FOAMI to maintain a balance between consensus and diversity.

Since the algorithm only performs a single forward calculation, it converges quickly and adds little computational overhead. The complexity grows with the number of models, but remains manageable in practical settings. Still, several limitations exist: it depends on manually tuned parameters (α , β) and may strengthen wrong predictions if multiple models are overconfident. Adding calibration steps or learnable weighting would help make the method more robust and scalable.

4 Experiments and Evaluation

4.1 Implementation Details

Training Settings: We use H2O Automated Machine Learning (AutoML) to automatically explore candidate models and tune their hyperparameters. In our experiments, AutoML discovered a total of 14 models within the given time and seed configuration, including multiple variants of tree-based learners (ExtraTrees,

Table 1. Performance comparison of augmentation methods on ICS dataset (%).

Method	XGBoost			ExtraTrees			RandomForest		
	Acc	F1	DR	Acc	F1	DR	Acc	F1	DR
Baseline	<u>83.88</u>	<u>83.12</u>	<u>83.88</u>	<u>85.11</u>	<u>84.64</u>	<u>85.11</u>	<u>84.86</u>	<u>84.31</u>	<u>84.86</u>
TVAE	<i>85.50</i>	<i>85.03</i>	<i>85.50</i>	85.55	85.17	85.55	85.50	85.00	85.50
SMOTE	<i>84.52</i>	<i>83.97</i>	<i>84.52</i>	<i>85.36</i>	<i>84.95</i>	<i>85.36</i>	84.62	84.09	84.62
GPT2	83.88	83.02	83.88	<i>85.26</i>	<i>84.76</i>	<i>85.26</i>	84.12	83.33	84.12
TabDiff	85.75	85.34	85.75	<i>85.21</i>	<i>84.69</i>	<i>85.21</i>	<i>85.06</i>	<i>84.49</i>	<i>85.06</i>
CopulaGAN	<i>84.17</i>	<i>83.22</i>	<i>84.17</i>	85.11	84.60	85.11	84.86	84.32	84.86
GaussianC	83.83	83.20	83.83	85.06	84.53	85.06	84.52	83.93	84.52
CTGAN	<i>84.22</i>	<i>83.61</i>	<i>84.22</i>	84.96	84.49	84.96	84.66	84.16	84.66

RandomForest, LightGBM, CatBoost, XGBoost), two neural networks (NeuralNetTorch and NeuralNetFastAI), two k-nearest neighbor models (KNeighborsDist and KNeighborsUnif), and a weighted ensemble meta-learner (WeightedEnsemble_L2). Based on validation performance, the GBT classifiers are selected for our study. For deep models, we employ BiLSTM and ResDNN implemented in TensorFlow and optimized using the Adam optimizer. Both models are trained with a learning rate of 0.001, batch size of 128, and 50 epochs, using early stopping with a patience of 5 based on validation F1-score. Dropout regularization (rate = 0.3) and batch normalization are applied to prevent overfitting. All models are trained and evaluated on a single NVIDIA T4 GPU (16 GB VRAM).

Evaluation Metrics: The detection performance is measured using confusion matrix metrics: Accuracy (Acc), Precision (Prec), F1-score (F1), Detection Rate (DR/Recall), and False Alarm Rate (FAR). Since IEC-104 is a multi-label classification task [7], we report macro-average values instead of micro-average to give equal importance to all classes.

4.2 Data Augmentation Evaluation

A central challenge in evaluating synthetic tabular data is the absence of a standardized metric for determining whether one generation method is superior to another, as evaluation remains inherently complex and an open research gap [20]. Existing approaches are usually categorized into two dimensions: *data availability* and *privacy protection* [20]. Data availability focuses on utility, including machine learning efficiency, fidelity to the source distribution, and alignment with domain knowledge, while privacy protection assesses the safeguarding of sensitive information using metrics such as Distance to Closest Record (DCR), attribute inference, or membership inference attacks.

In our setting, privacy protection is not a concern since the ICS datasets contain no personally identifiable information. Likewise, alignment with human knowledge is not applicable, as our data does not involve domain-level semantic constraints beyond the observed network traffic patterns. Fidelity reflects how well synthetic data preserves the statistical properties of the original distribution, but high fidelity does not necessarily imply strong downstream performance.

Table 2. Ablation Studies on IEC-104 (%).

Study	Method	Acc	Prec	F1	DR	FAR
S1	XGB	83.88	83.05	83.12	83.88	1.47
	ET	85.11	84.87	84.64	85.11	1.35
	RF	84.86	84.58	84.31	84.86	1.38
	GBM	83.97	83.38	83.42	83.97	1.45
	CB	83.48	82.58	82.65	83.48	1.50
S2	XGB	85.26 ^{↑1.38}	84.87 ^{↑1.82}	84.77 ^{↑1.65}	85.26 ^{↑1.38}	1.34 ^{↓0.13}
	ET	85.55 ^{↑0.44}	85.37 ^{↑0.50}	85.18 ^{↑0.54}	85.55 ^{↑0.44}	1.32 ^{↓0.03}
	RF	85.40 ^{↑0.54}	85.21 ^{↑0.63}	84.93 ^{↑0.62}	85.40 ^{↑0.54}	1.33 ^{↓0.05}
	GBM	84.51 ^{↑0.54}	84.14 ^{↑0.76}	84.06 ^{↑0.64}	84.52 ^{↑0.55}	1.41 ^{↓0.04}
	CB	83.58 ^{↑0.10}	82.62 ^{↑0.04}	82.73 ^{↑0.08}	83.58 ^{↑0.10}	1.49 ^{↓0.01}
S3	XGB	86.69 ^{↑2.81}	86.49 ^{↑3.44}	86.27 ^{↑3.15}	86.69 ^{↑2.81}	1.21 ^{↓0.26}
	ET	86.64 ^{↑1.53}	86.61 ^{↑1.74}	86.30 ^{↑1.66}	86.64 ^{↑1.53}	1.21 ^{↓0.14}
	RF	86.74 ^{↑1.88}	86.81 ^{↑2.23}	86.46 ^{↑2.15}	86.74 ^{↑1.88}	1.21 ^{↓0.17}
	GBM	85.75 ^{↑1.78}	85.25 ^{↑1.87}	85.21 ^{↑1.79}	85.75 ^{↑1.78}	1.30 ^{↓0.15}
	CB	87.08 ^{↑3.60}	87.00 ^{↑4.42}	86.81 ^{↑4.16}	87.08 ^{↑3.60}	1.17 ^{↓0.33}
	ResDNN	69.97	70.49	68.70	69.97	2.73
	BiLSTM	68.15	66.99	66.68	68.15	2.89
S4	HVE	88.21 ^{↑3.10}	88.22 ^{↑3.35}	88.04 ^{↑3.40}	88.21 ^{↑3.10}	1.07 ^{↓0.28}
	WVE	88.95 ^{↑3.84}	89.13 ^{↑4.26}	88.80 ^{↑4.16}	88.95 ^{↑3.84}	1.00 ^{↓0.35}
	MI	89.00 ^{↑3.89}	89.18 ^{↑4.31}	88.85 ^{↑4.21}	89.00 ^{↑3.89}	0.99 ^{↓0.36}

Therefore, our evaluation focuses on the practical utility of synthetic data in intrusion detection tasks.

Table 1 reports the performance of the different augmentation methods when applied to our dataset. Each method was configured to generate a synthetic dataset of the same size as the original training data. From these results, TVAE consistently achieved the highest improvement in classification metrics, and was therefore selected as our augmentation method of choice.

After selecting TVAE, we applied it to generate additional data for our main experiments. Error analysis showed that classes 7, 8, 9, and 10 were consistently harder to classify and more prone to confusion with other categories. To address this weakness, we applied targeted augmentation by generating 800 additional synthetic samples for each of classes 7, 8, 9, and 10, and 200 synthetic samples for each of the remaining classes. The purpose of this strategy was to increase data diversity in underperforming categories, strengthen inter-class boundaries, and reduce misclassification among overlapping attack types.

4.3 Ablation Studies

To better understand the role of each building block in our framework, we performed ablation experiments on the IEC-104 dataset. The study was divided into four incremental settings:

- **S1** – *Baseline performance*: We first established baselines using five standard learners (XGB, ET, RF, GBM, and CB) trained on the raw feature set.

- **S2** – *Applying CSFS*: The same models were then retrained after integrating CSFS, which filters out features that exhibit instability under distributional shifts or contribute little predictive information.
- **S3** – *Impact of augmentation*: Models trained on CSFS-refined features with TVAE-generated synthetic data. In addition to GBT models, two DL architectures (BiLSTM and ResDNN) are trained on this data.
- **S4** – *Ensemble strategies*: Three fusion methods are applied to combine GBT and DL models, including Hard Voting Ensemble (HVE), Weighted Voting Ensemble (WVE), and Mutual Inference (MI).

The comprehensive experimental findings from these four scenarios are encapsulated in [Table 2](#) and elaborated upon below:

S1 results: On the IEC-104 dataset, the tree-based models outperform the boosting methods. ET has the best result with $F1 = 84.64\%$ and $FAR = 1.35\%$, while CB has the worst result with $F1 = 82.65\%$ and $FAR = 1.50\%$. This result will be used as a baseline for comparison with the following methods.

S2 results: The application of CSFS resulted in consistent improvements across all models. For example, XGB improved from $F1 = 83.12\%$ to 84.77% , while reducing FAR from 1.47% to 1.34% . From the above results, it can be asserted that **RQ1** has been addressed, as CSFS improved the training quality by removing unstable features and reducing drift in the dependent variable, thereby improving the recognition performance.

S3 results: In the augmentation phase, synthetic data were generated with an uneven allocation: 800 samples for classes 7, 8, 9, 10 (the most frequently confused) and 200 for all other classes. This targeted strategy substantially improved recognition of difficult groups. CB achieved the best GBT performance with $F1 = 86.81\%$ and $FAR = 1.17\%$.

Although CB after augmentation has good improvement results, there are still many shortcomings in recognizing layers 7, 8, 9, and 10. When AutoML is used to select the base model, DL models are not highly evaluated, but we still use two DL models, BiLSTM and ResDNN, to train on the augmentation dataset. Although their overall $F1$ scores are lower (around 68–70%), both show better ability to distinguish layers 7 and 10 than CB, as shown in [Fig. 2](#). This emphasizes that deep architectures can exploit augmented data to reduce layer-by-layer confusion, even if their aggregate performance is worse. The observed improvements confirm that **RQ2** has been addressed. Augmentation benefits both GBTs (improving overall $F1$ and FAR) and DL models (enhancing separation of hard-to-distinguish classes).

S4 results: Finally, the ensemble strategies were applied to combine the GBT and DL models in [Table 2](#) with HVE achieving $F1 = 88.04\%$ and $FAR = 1.07\%$, WVE with the model weights of CB (0.25), RF (0.35), BiLSTM (0.10) and ResDNN (0.30) achieving $F1 = 88.80\%$ with $FAR = 1.00\%$, and MI giving the best results with $F1 = 88.85\%$ and $FAR = 0.99\%$. These results demonstrate that the ensembles take advantage of the complementary strengths of GBT and DL models. The combination of them provides good overall accuracy and avoids false alarms. Thus, **RQ3** being addressed, the ensemble strategies, especially

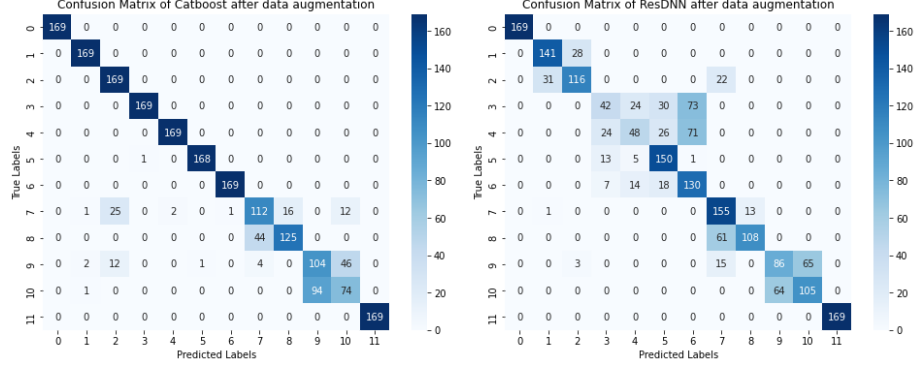


Fig. 2. Confusion matrices of CatBoost and ResDNN after augmentation.

Table 3. Comparison again SoTA approaches on the IEC-104 benchmark (%).

Method	Acc(%)	Prec(%)	F1(%)	DR(%)	FAR(%)
FOAMI (our)	89.00	89.18	88.85	89.00	0.99
CSFS [16]	85.60	85.40	85.22	85.60	1.31
TDAELID [22]	85.44	85.49	84.88	85.44	1.31
TS+CART [17]	83.14	-	82.58	83.14	1.53
FGSM+CTGAN [3]	82.44	-	80.98	82.44	1.59

MI combining GBT and DL have provided the most effective and sustainable approach to improve detection accuracy and ensure adaptability to evolving ICS threats.

4.4 Comparison to SoTA Methods

To evaluate the efficacy of FOAMI, we performed a thorough comparison to recent methods already used on the standard IEC-104 benchmark. Table 3 indicates that FOAMI far outperforms SOTA approaches in the majority of evaluation metrics. FOAMI attained Acc of 89.00%, an F1 of 88.85%, and a DR of 89.00%, while maintaining the lowest FAR at 0.99%. Meanwhile, methods like CSFS [16] and TDAELID [22] only achieve a DR of approximately 85% and have a higher FAR (1.31%), indicating lower reliability in real-world deployment environments. Notably, data augmentation methods such as FGSM+CTGAN [3] performed worse, with an Acc of only 82.44% and an F1 dropping to 80.98%. These results confirm FOAMI’s ability to learn distinctive features and minimize false alarms, thanks to the effective combination of feature optimization, realistic data augmentation, and mutual inference. Experimental evidence indicates that FOAMI is a powerful and highly scalable solution for detecting threats to ICS.

5 Conclusion

This paper proposes the FOAMI framework based on feature set optimization, realistic data augmentation, and mutual inference to improve the performance and

reliability of threat detection systems in ICS networks. FOAMI picks the best set of features using the CSFS method, which mixes important weights for changes in data distribution with the clarity provided by SHAP. To address the challenges of limited sample size and the quality of the training set, we developed a realistic data augmentation strategy, evaluated various SoTA methods, and selected the TVAE technique for data augmentation. To take advantage of the strengths of GBT and DL models, we proposed a method where they work together to check and confirm each other’s results, which helps lower mistakes and find unusual behaviors that one model alone might overlook. Our approach addresses the root causes of model degradation by shifting the distribution and reducing confusion between attack types. Deep testing on the ICS dataset involving the IEC-104 protocol has demonstrated that this framework outperforms advanced methods, maintaining strong detection capabilities and a low false alarm rate.

Nonetheless, FOAMI has several limitations, having examined just one experimental ICS protocol and not yet assessed other feature extraction methods from network traffic, including those from CICFlowmeter. The validation of freshly produced samples mostly depends on performance indicators including F1, DR, and FAR. In the future, these constraints will propel us to further investigate and enhance the FOAMI method, including conducting additional assessments of other ICS protocols such as DNP3, PROFINET, or Modbus/TCP; integrating reinforcement learning mechanisms with MI and so forth.

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